BinFoLab

Density Functional Theory

A beginner's guide to the theory, its applications, and every step of the DFT-Flow pipeline

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For students and researchers in bioinformatics, pharmacy, chemistry and the life sciences who are new to computational quantum chemistry. No prior knowledge of quantum mechanics is assumed.

Companion to the **BinFoLab DFT-Flow** Google Colab notebook (v1.0)

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1. How to use this guide

Density functional theory (DFT) is now a routine part of papers in medicinal chemistry, natural products, pharmacology and materials science. Many researchers meet it for the first time as a figure of coloured orbitals or a table of “HOMO–LUMO gaps” in someone else's paper, without knowing what was computed or why. This guide explains both.

Part	What it covers	Read it when
A. Understanding DFT (Sections 2–5)	What DFT is, the vocabulary (functional, basis set, SCF, dispersion, solvation), what it is used for, and its limits	Before your first calculation
B. The DFT-Flow pipeline (Sections 6–17)	Each stage of the notebook: what it does, why it is needed, what it produces and what to check	While running the notebook
C. Using results well (Sections 18–21)	Interpreting numbers, reporting in a manuscript, common mistakes, glossary, further reading	When writing up

> **BEGINNER TIP**
> Read Sections 2 and 3 once, even if you skip the equations. Every later step builds on three ideas introduced there: the **electron density**, the **functional**, and the **basis set**.

2. Why do we need quantum chemistry?

A molecule is a set of positively charged nuclei held together by a cloud of negatively charged electrons. Almost everything a chemist or biologist cares about is decided by those electrons: the molecule's shape, which bonds break, where it is attacked by an enzyme, how it binds a receptor, what colour it is, and how it absorbs infrared light.

Most bioinformatics tools do not model electrons at all. Molecular docking and molecular dynamics use **force fields**, which treat atoms as balls connected by springs with fixed partial charges. Force fields are fast enough to simulate a protein for microseconds, but they cannot describe anything that depends on how electrons rearrange: chemical reactions, charge transfer, orbital energies, or spectra.

Quantum chemistry models the electrons directly. In principle, everything follows from the Schrödinger equation:

$$\hat{H}\,\Psi = E\,\Psi$$

Here $\hat{H}$ is the energy operator (kinetic energy plus all attractions and repulsions), Ψ is the **wavefunction**, and E is the energy. The difficulty is that Ψ depends on the coordinates of every electron at once. Phenol (C_6H_5OH) has 50 electrons, so its wavefunction is a function of 150 spatial coordinates. Solving this exactly is impossible for anything larger than a few electrons, and accurate approximations to the wavefunction become extremely expensive as molecules grow.

> **KEY IDEA**
> DFT avoids the many-electron wavefunction. Instead it works with the **electron density** $\rho(r)$, which depends on only three coordinates (x, y, z) however many electrons the molecule has. This is why DFT can treat drug-sized molecules at a cost that ordinary computers, and now free Colab GPUs, can afford.

3. What is density functional theory?

3.1 The core idea: the density decides everything

In 1964 Hohenberg and Kohn proved two theorems that underpin DFT. The first says that the ground-state electron density uniquely determines the molecule's energy and all its ground-state properties. The second says that the true density is the one that gives the lowest energy, so you can find it by minimisation. A quantity that takes a whole function (here the density) and returns a number (the energy) is called a **functional**, written E[ρ]. That is where the name "density functional theory" comes from. Walter Kohn shared the 1998 Nobel Prize in Chemistry for this work.

3.2 The Kohn–Sham method

The theorems prove that an energy functional exists but do not say what it is. In 1965 Kohn and Sham made DFT practical: they replaced the real, interacting electrons with an imaginary system of non-interacting electrons that has exactly the same density. These imaginary electrons occupy **Kohn–Sham orbitals**, which play the same role as the familiar molecular orbitals of chemistry textbooks. The total energy is split into parts:

$$E[\rho] = T_s[\rho] + E_{ne}[\rho] + J[\rho] + E_{xc}[\rho]$$

Term	Meaning	Known exactly?
T_s	Kinetic energy of the non-interacting electrons	Yes
E_{ne}	Attraction between electrons and nuclei	Yes
J	Classical repulsion of the electron cloud with itself	Yes
E_{xc}	Exchange–correlation energy: every remaining quantum effect (the Pauli principle, electrons avoiding each other, and the correction to the kinetic energy)	No — must be approximated

All the difficulty of quantum mechanics is pushed into the single term E_{xc}. Different DFT methods differ only in how they approximate it. That approximation is what people mean when they say "the functional" (B3LYP, PBE0, M06-2X and so on).

3.3 The self-consistent field (SCF) loop

The orbitals depend on the potential the electrons feel, but that potential depends on the density, which is built from the orbitals. DFT programs break this circle by iteration:

1. Guess an initial density (usually a sum of free-atom densities).
2. Build the effective potential from that density, including the exchange–correlation part.
3. Solve the Kohn–Sham equations to get new orbitals.
4. Build a new density from the occupied orbitals.
5. Compare with the previous density. If the energy and density no longer change (within a threshold, e.g. 10^{-9} hartree), stop. Otherwise return to step 2.

When the loop stops, the calculation is said to be **converged**. Every number DFT reports comes from a converged SCF; an unconverged one is meaningless. DFT-Flow checks convergence for every calculation and applies rescue strategies (level shifting, damping, a second-order solver) when the standard procedure struggles.

3.4 Choosing a functional: Jacob's ladder

Functionals are often arranged on “Jacob's ladder” (a metaphor due to John Perdew). Each rung adds a new ingredient, usually improving accuracy at extra cost.

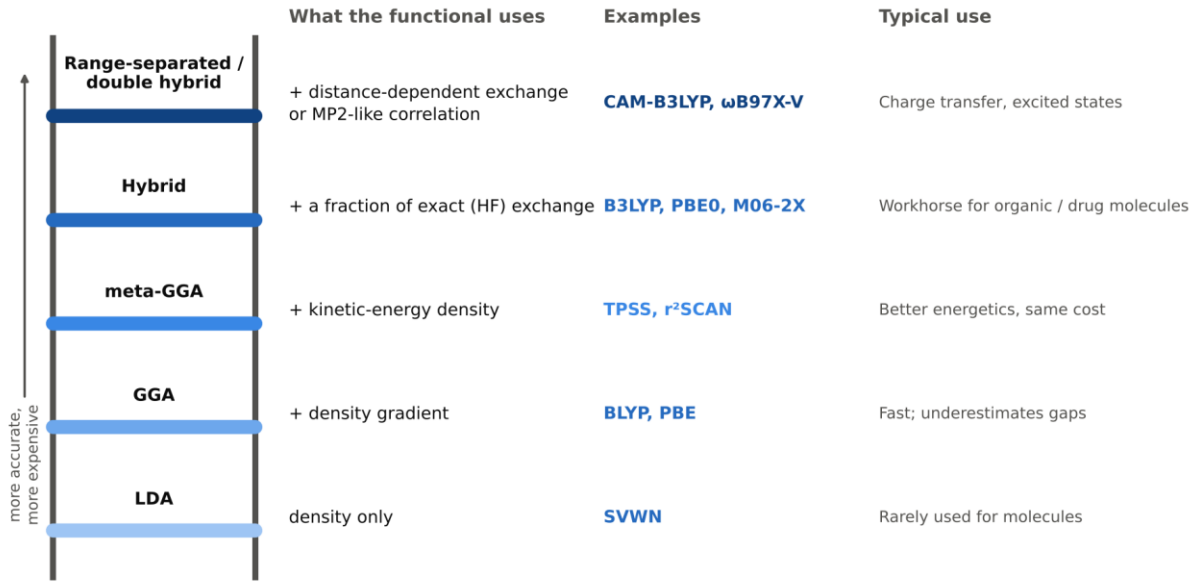


Figure 1. Jacob's ladder of exchange–correlation functionals, with representative members and typical uses.

Hybrid functionals mix in a fraction of “exact” exchange from Hartree–Fock theory. **B3LYP**, which contains 20 % exact exchange, has been the most widely used functional in chemistry for three decades. It is reliable for the geometries, vibrational frequencies and orbital trends of organic molecules, which is why most drug-design papers use it. **Range-separated hybrids** such as CAM-B3LYP and ωB97X-V increase the exact exchange at long electron–electron distances. This corrects a known failure of ordinary functionals for charge-transfer excited states, so they are preferred for UV-Vis spectra (TD-DFT).

> **BEGINNER TIP**
> There is no single “best” functional. Use one that is well established for your type of molecule and property, state it clearly, and compare molecules only when they were calculated with the same functional.

3.5 Dispersion corrections

Ordinary functionals miss **London dispersion**, the weak attraction between temporarily polarised electron clouds. It is responsible for π-stacking, the folding of flexible molecules onto themselves, and much of the binding of a ligand in a protein pocket. Grimme's D3 and D4 corrections add a cheap atom-pair term that restores it. Modern best practice is to always use a dispersion correction;

you will see it written as a suffix, for example B3LYP-D3(BJ), where BJ denotes the Becke–Johnson damping function.

3.6 Basis sets

Orbitals are built from a set of mathematical functions centred on each atom, called the **basis set**. More functions give more flexibility and a better answer, but cost rises steeply. The common families have a naming code:

Basis set	Quality	Typical use	Relative cost
def2-SVP, 6-31G(d)	Double-ζ with polarisation (2 functions per valence orbital)	Geometry optimisation and frequencies	Low
def2-TZVP, 6-311G(d,p)	Triple-ζ with polarisation	Energies, orbitals, dipoles, final properties	Medium
def2-TZVPP, def2-QZVP	Triple/quadruple-ζ with extra polarisation	High-accuracy benchmarks	High
"+" or "aug-" added (e.g. 6-311++G(d,p))	Adds diffuse functions	Anions, excited states, weak interactions	Higher

Polarisation functions (the "P" in SVP or "(d)" in 6-31G(d)) let orbitals change shape when bonds form; they should always be included. The **def2** family covers the whole periodic table up to radon and automatically uses effective core potentials for heavy elements such as iodine, so it is the default in DFT-Flow.

3.7 Solvation

Calculations in the "gas phase" treat the molecule as isolated in vacuum. Biology happens in water. **Implicit solvation models** such as the polarizable continuum model (PCM) surround the molecule with a continuous dielectric medium that responds to its charge distribution. This is cheap and captures most of the bulk electrostatic effect of the solvent, though not specific hydrogen bonds to individual water molecules.

Use the gas phase when…	Use a solvent model when…
comparing with gas-phase experimental data (e.g. ionisation energies) reproducing literature that used gas phase screening many molecules quickly	studying charged or zwitterionic species interpreting solution UV-Vis spectra or pKa relating results to biological or assay conditions

3.8 Reading the level-of-theory notation

Papers summarise the method in one compact string. DFT-Flow prints it for you, for example:

$$\text{B3LYP-D3(BJ)/def2-TZVP // B3LYP-D3(BJ)/def2-SVP}$$

Read it from right to left: the geometry was optimised with B3LYP plus D3(BJ) dispersion and the def2-SVP basis (after the "//"), and the final energies and properties were then computed on that

geometry with the larger def2-TZVP basis (before the "//"). Geometries converge quickly with basis-set size while energies converge slowly, so this split gives most of the accuracy for a fraction of the cost.

3.9 How expensive is DFT?

The cost of hybrid DFT grows roughly with the third to fourth power of molecular size, so doubling the number of atoms can make a calculation 8–16 times slower. A practical measure of size is the number of **basis functions**, which DFT-Flow reports for each molecule. Graphics processors (GPUs) accelerate DFT dramatically; with the GPU4PySCF library, single points and optimisations of typical drug-sized molecules (roughly 20–60 atoms) are practical on a free Colab GPU, while the same work on a laptop CPU can take hours.

4. What is DFT used for?

DFT is used wherever electronic structure matters. The table below maps common research questions to the DFT quantity that answers them.

Field	Typical question	DFT quantity used
Drug discovery	Which atoms of a hit compound are reactive or metabolically labile? How do charges influence binding?	Fukui functions, MEP, atomic charges, HOMO/LUMO
Molecular docking & MD	Is the ligand geometry and charge model realistic?	Optimised geometry, ESP-derived charges
Natural products & antioxidants	Which phenol is the better radical scavenger, and by which mechanism?	Ionisation energy, bond dissociation enthalpy, ω^-
Spectroscopy	Which band in the FT-IR or UV-Vis spectrum belongs to which group or transition?	Vibrational frequencies, TD-DFT excitations
Structure assignment	Which isomer or tautomer is the most stable?	Relative Gibbs free energies
Reaction mechanisms & catalysis	What is the rate-limiting step? Which pathway is preferred?	Transition states, activation barriers
Materials & nanoscience	Band gaps, adsorption of molecules on surfaces, nanozyme active sites	Periodic DFT (different software, e.g. Quantum ESPRESSO)
Corrosion inhibitors, sensors, dyes	Will a molecule donate electrons to a metal? Which dye absorbs at the right wavelength?	HOMO/LUMO energies, ΔN, TD-DFT

4.1 DFT in a typical computer-aided drug-design paper

In bioinformatics and medicinal-chemistry papers, DFT usually appears as one layer of a larger workflow:

1. **Virtual screening or de novo design** proposes candidate molecules.
2. **Molecular docking** ranks how well they fit the target pocket.
3. **Molecular dynamics** checks whether the best complexes are stable over time.
4. **DFT** characterises the electronic properties of the top hits: reactivity (HOMO–LUMO gap, Fukui functions), electrostatic complementarity (MEP), and stability (frequencies).
5. **ADMET prediction** estimates absorption, metabolism and toxicity.

DFT adds a physically grounded explanation of *why* a compound might bind or react the way it does. It does not by itself predict biological activity, and good papers say so.

5. Strengths and limitations of DFT

Strengths	Limitations
Treats electrons explicitly, so it describes bonding, charge distribution and reactivity	Results depend on the chosen functional; there is no systematic way to improve it
Good accuracy for cost: geometries within about 0.01–0.02 Å, frequencies within a few percent	Self-interaction error: HOMO–LUMO gaps from orbital energies are too small, and charge-transfer states are poorly described by simple functionals
Gives spectra (IR, UV-Vis, NMR) that can be compared directly with experiment	Misses dispersion unless a correction is added
Applicable to drug-sized molecules on modest hardware, especially GPUs	Usually one conformer at 0 K with implicit solvent; real molecules are dynamic and solvated
Mature, well-validated methods with large literature for comparison	Struggles with strongly correlated systems (some transition-metal complexes, diradicals)

WATCH OUT

DFT numbers are most reliable as **relative** quantities. Use them to compare molecules computed at the same level of theory, or to compare with experiment after validation. An isolated absolute value, such as “the HOMO is −6.2 eV”, has limited meaning on its own.

6. The DFT-Flow pipeline at a glance

DFT-Flow is a Google Colab notebook that runs the complete protocol described above, from a SMILES string to publication figures. It is built on the open-source PySCF program with GPU4PySCF acceleration, so no commercial licence is required. Figure 2 shows the ten stages.

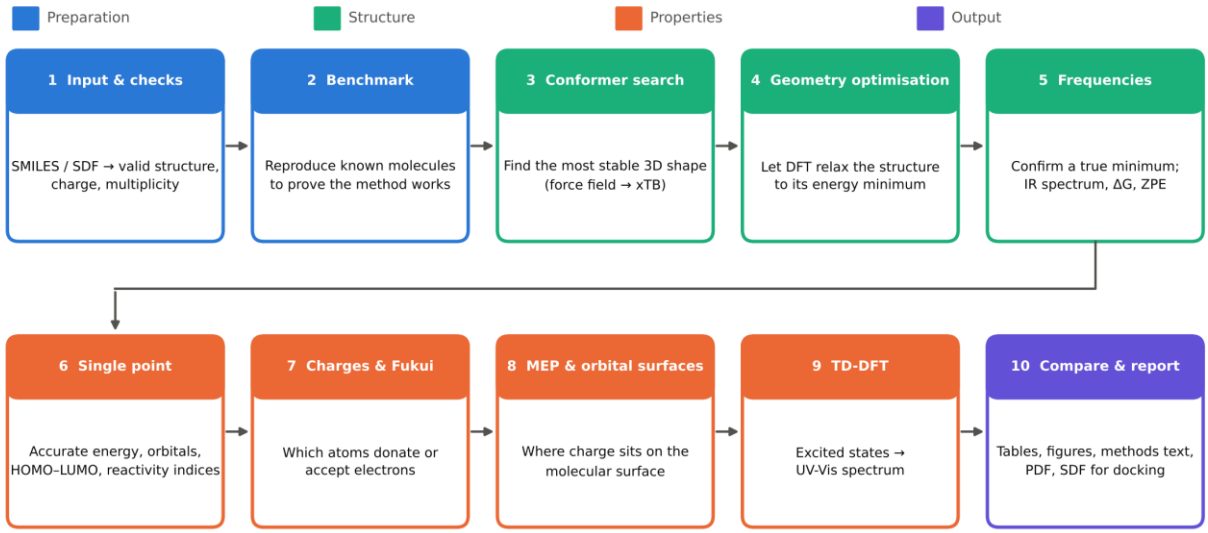


Figure 2. The DFT-Flow workflow. Preparation stages check the input and the method; structure stages find a reliable geometry; property stages compute the electronic descriptors; the output stage compiles tables, figures and the report.

Stage	Notebook part	Main question it answers
Setup & settings	Parts 1–2	Is the software installed and which level of theory will be used?
1. Input & checks	Part 3	Are the structures, charges and spin states valid?
Engine & 2. Benchmark	Parts 4–5	Does the method reproduce known experimental values?
3. Conformer search	Part 6	What is the most stable 3D shape?
4. Geometry optimisation	Part 7	What is the exact DFT equilibrium structure?
5. Frequencies	Part 8	Is it a true minimum? What are its IR spectrum and free energy?
6. Single point	Part 9	How reactive is the molecule overall?
7. Charges & Fukui	Part 10	Which atoms are the reactive sites?
8. Surfaces	Part 11	Where are the electron-rich and electron-poor regions?
9. TD-DFT	Part 12	How does the molecule absorb light?
10. Compare & report	Parts 13–14	How do the molecules rank, and how do I report it?

EXAMPLE

> The figures in this guide come from a small test run of the notebook on formamide, acetic acid and phenol at the B3LYP-D3(BJ)/def2-SVP level. They show what each output looks like; your own molecules will give their own values.

6.1 Checkpoints and resuming

Free Colab sessions can disconnect. DFT-Flow saves every finished stage for every molecule, and saves the geometry after each optimisation cycle. If the session ends, simply run the notebook again from the top: completed work is skipped automatically and an interrupted optimisation continues from its last geometry. For long runs, tick **Save to Google Drive** so the checkpoints survive the session.

7. Setup and settings (Parts 1–2)

What it does

Part 1 installs the software (PySCF, geomeTRIC, xTB, RDKit and plotting tools) and detects whether a GPU is available. Part 2 is a form where you choose the level of theory, solvent and which analyses to run.

Why it is needed

The level of theory determines every number the notebook produces. Fixing it in one place, and printing it as a single notation string, keeps the whole study consistent and makes it reproducible.

The presets

Preset	Geometry & frequencies	Single point & properties	When to use
Fast screening	B3LYP-D3(BJ)/def2-SVP	B3LYP-D3(BJ)/def2-SVP	Many molecules; first look
Standard (publication)	B3LYP-D3(BJ)/def2-SVP	B3LYP-D3(BJ)/def2-TZVP	Most papers on drug-like molecules
High accuracy	PBE0-D3(BJ)/def2-TZVP	ωB97X-V/def2-TZVPP	Small sets where accuracy matters; needs a strong GPU

What to check

- The backend line says **GPU (GPU4PySCF)**. If it says CPU, change the runtime type to GPU unless your molecules are very small.
- Copy the printed **level of theory** string into your notes; you will need it for the methods section.
- If you change the level of theory later, the notebook archives old results and starts fresh, so results at different levels are never mixed.

8. Step 1 — Input molecules and validation (Part 3)

8.1 Providing and checking molecules

What it does

You provide molecules as SMILES strings (in a pasted table or a CSV file) or as 3D structures (SDF, MOL or XYZ). The notebook adds hydrogens, reads the total charge from the formal charges, determines the spin multiplicity, checks that every element is covered by the chosen basis sets, counts basis functions to estimate cost, and draws a 2D grid of the structures.

Why it is needed

Errors in the input are the most common cause of wrong DFT results, and they are silent: DFT will happily compute the wrong molecule. Two quantities must be right before anything else:

Charge. A carboxylic acid (–COOH) and its carboxylate ($–COO^-$) are different molecules with different properties. DFT never changes protonation for you, so supply the form you want to study, for example the form dominant at pH 7.4.

Spin multiplicity (2S+1). Most stable organic molecules have all electrons paired: a singlet, multiplicity 1. Radicals have one unpaired electron: a doublet, multiplicity 2. The number of electrons and the multiplicity must be compatible (an even electron count needs an odd multiplicity), and the notebook rejects impossible combinations.

What you get

- A validation table: formula, molecular weight, atom counts, charge, multiplicity, basis-function counts, a cost tier, and a status for each molecule.
- A 2D structure grid to confirm the molecules are the ones you intended.
- A template file (sample_input_template.csv) showing the expected input format.

What to check

- Every molecule you intended is marked **ok**; read the notes column for corrected charges.
- Look at the 2D grid: wrong stereochemistry or a missing ring in a SMILES string is easy to spot here.
- “Heavy” or “Very heavy” cost tiers on a CPU runtime signal very long run times.

9. Step 2 — The DFT engine and method validation (Parts 4–5)

9.1 Engine set-up and smoke test

What it does

Part 4 defines how each calculation is built (functional, dispersion, integration grid, density fitting, solvent) and runs a quick calculation on water to prove the setup works on your hardware. If the GPU fails, the notebook switches to CPU here rather than halfway through your molecules.

Why it is needed

A failure in the very first second is far cheaper than a failure after an hour of optimisation.

What you get

- A “Smoke test passed” message with the energy of water and the time taken.

What to check

- The energy is close to −76.3 hartree (the exact value depends on functional and basis).

9.2 Benchmark against experiment

What it does

The notebook optimises three small reference molecules (water, formaldehyde, ammonia) at your chosen level and compares bond lengths, angles and dipole moments with experimental values. Each quantity passes if it is within a tolerance typical of hybrid DFT: ±0.025 Å for bonds, ±2.5° for angles, ±0.25 D for dipoles.

Why it is needed

Validation shows that your protocol reproduces reality before you apply it to unknown molecules. A large deviation reveals a setup problem; a small, systematic one tells you the method's known bias. Reporting this table in the supporting information also reassures reviewers that the protocol was checked.

What you get

- A table of computed versus experimental values with PASS/FAIL, and a deviation chart (Figure 3).

What to check

- Most quantities should pass. A small dipole failure with def2-SVP is expected, because dipole moments need a triple-ζ basis; the Standard preset uses one for the single point.

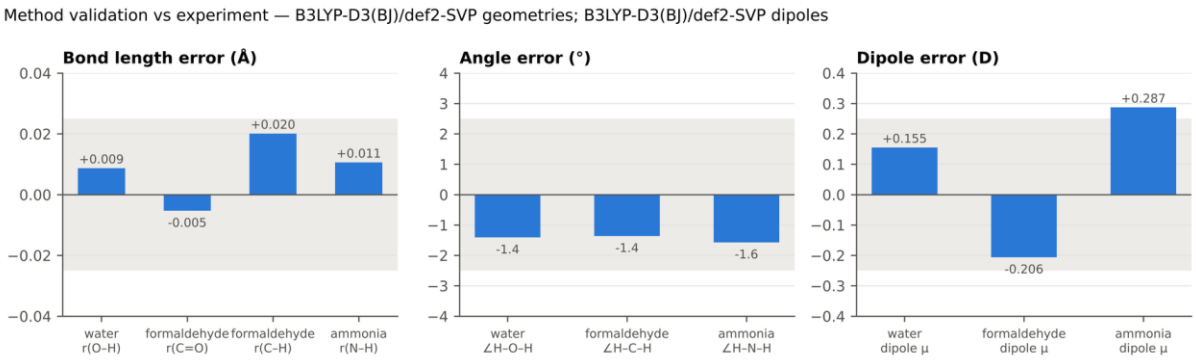


Figure 3. Example benchmark at B3LYP-D3(BJ)/def2-SVP. Bars show computed minus experimental values; the grey band is the tolerance. The ammonia dipole lies just outside it, the expected def2-SVP basis-set effect.

10. Step 3 — Conformer search (Part 6)

10.1 Finding the most stable 3D shape

What it does

Flexible molecules can adopt many 3D shapes (**conformers**) by rotating around single bonds. The notebook finds a good one with a three-level "funnel", from cheap to accurate:

ETKDGv3 (RDKit) generates many diverse 3D structures from the 2D graph using distance geometry and known torsion preferences. **MMFF94**, a force field, relaxes each one in milliseconds. **GFN2-xTB**, a fast semi-empirical quantum method, re-optimises the best few and ranks them far more reliably than the force field. The lowest-energy structure whose bonding is unchanged is passed to DFT.

Why it is needed

DFT optimisation only finds the nearest minimum to where it starts, so it must start from a sensible conformer. Properties such as the dipole moment, MEP and even the HOMO–LUMO gap can change between conformers, especially when intramolecular hydrogen bonds form or break. Doing the search with DFT directly would cost hundreds of DFT optimisations; the funnel spends DFT effort only on the best candidate.

The relative populations of conformers follow the Boltzmann distribution: at room temperature, a conformer 1.36 kcal/mol higher in energy is about 10 times less populated than the lowest one.

What you get

- Relative xTB energies and Boltzmann populations of the refined conformers (Figure 4).
- The selected structure, saved as an XYZ file.

What to check

- The population of the selected conformer. Above about 0.7 one shape dominates and single-conformer results are representative. Below about 0.3 the molecule is genuinely flexible; mention this in your paper or compute a second conformer.
- The connectivity check: conformers in which xTB moved a proton or broke a bond are rejected automatically.

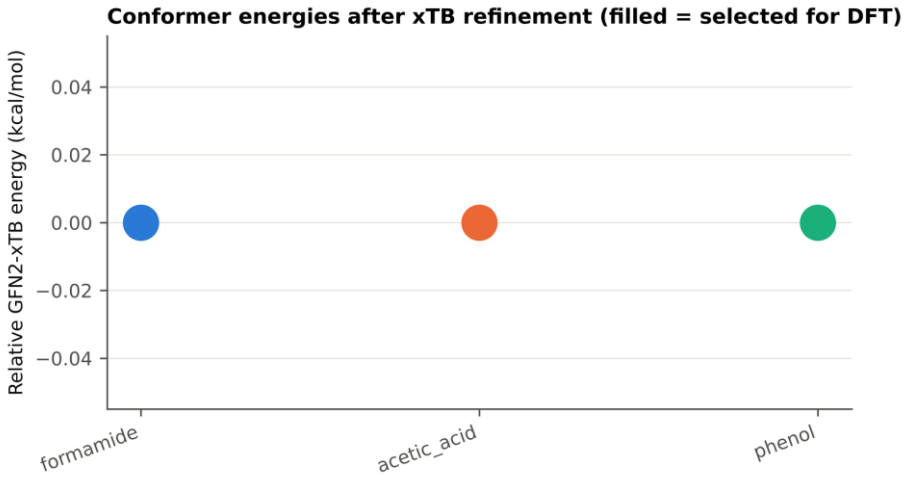


Figure 4. Conformer energies after xTB refinement. These three small, rigid test molecules each have a single conformer, so all points sit at 0 kcal/mol; flexible drug molecules show a spread of points.

11. Step 4 — DFT geometry optimisation (Part 7)

11.1 Relaxing the structure

What it does

Starting from the xTB structure, the notebook moves the atoms step by step downhill on the DFT **potential energy surface** until the forces on all atoms vanish. Think of a ball rolling into the bottom of a valley: at each step DFT computes the energy and its gradient (the force), and the optimiser (geomeTRIC) chooses the next geometry. Convergence requires the forces and atomic displacements to fall below standard thresholds equivalent to those in Gaussian.

Why it is needed

Every property depends on geometry. Bond lengths from force fields or xTB differ slightly from DFT, and orbital energies, charges and frequencies are only meaningful at the DFT minimum of the same method. Frequencies in particular are invalid at a non-stationary geometry.

What you get

- Optimised coordinates (XYZ), final energy, the number of cycles, and the RMSD between the xTB and DFT structures.
- Convergence traces of energy and gradient (Figure 5).

What to check

- **Converged = True** for every molecule. If not, re-run the cell: it continues from the last geometry.
- An RMSD below about 0.3 Å means xTB was already close. A large RMSD means the molecule rearranged, which is worth inspecting in the 3D viewer.

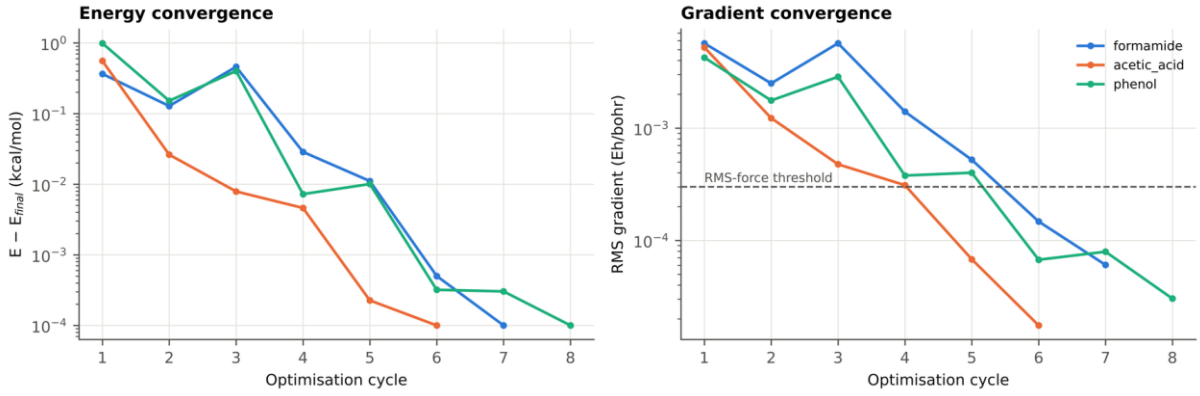


Figure 5. Optimisation convergence. Left: energy above the final value falls smoothly on a log scale. Right: the RMS gradient drops below the convergence threshold (dashed line).

12. Step 5 — Vibrational frequencies and thermochemistry (Part 8)

12.1 Confirming the minimum and computing spectra

What it does

The notebook computes the **Hessian**, the matrix of second derivatives of the energy with respect to atomic positions, and from it the 3N−6 vibrational frequencies of a molecule with N atoms. From the frequencies it calculates zero-point vibrational energy (ZPE), enthalpy (H), entropy (S) and Gibbs free energy (G) at the chosen temperature, and analytic IR intensities to simulate the IR spectrum.

Why it is needed

Checking the minimum. At a true minimum all frequencies are real (positive). An **imaginary frequency** (reported as a negative number) means the structure sits on a saddle point, a ridge rather than a valley, and is not a stable molecule. The notebook detects this, nudges the structure along the offending vibration and re-optimises automatically.

Thermochemistry. Chemical equilibria and reaction feasibility are decided by free energies, not raw electronic energies. ΔG between two tautomers or conformers tells you which dominates.

IR spectra. Calculated bands help assign experimental FT-IR spectra: which peak is the C=O stretch, which the O–H stretch.

What you get

- A thermochemistry table: ZPE, H, G, S, heat capacity, number of imaginary modes, lowest frequency.
- Simulated IR spectra with the strongest bands labelled (Figure 6), and a table of all bands.

What to check

- **n imag = 0** for every structure. Tiny imaginary modes (above −30i cm^{-1}) are usually numerical noise in very flat rotations.
- Harmonic DFT frequencies are typically a few percent higher than experiment. Before comparing with FT-IR, apply the scale factor for your functional and basis from the NIST CCCBDB database (set it in Part 2).
- Use differences in G between species, never an absolute G on its own.

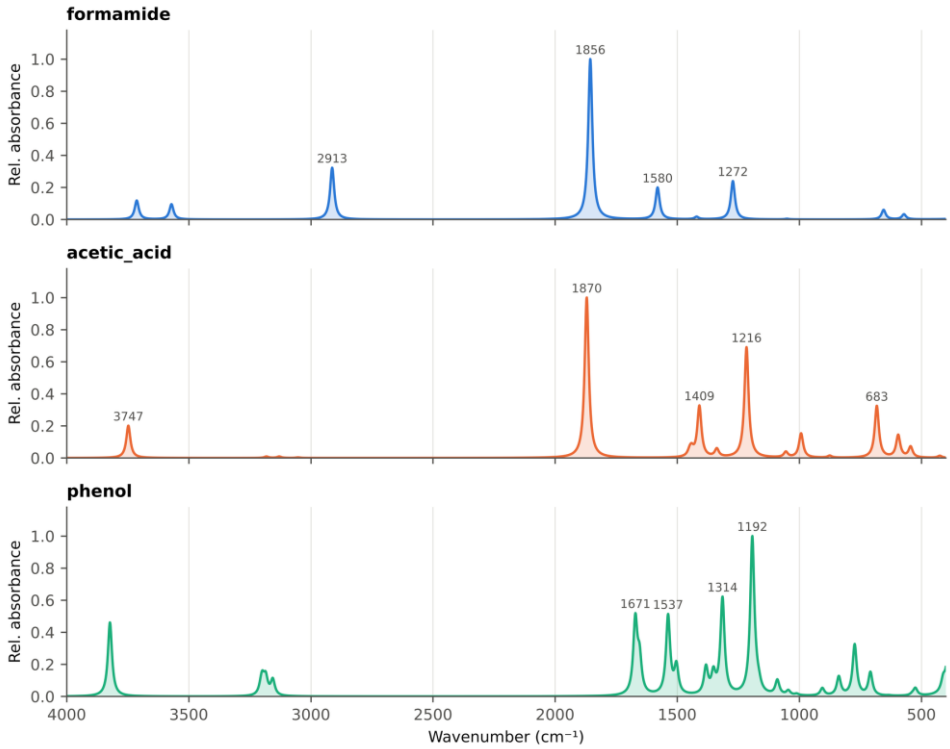


Figure 6. Simulated IR spectra (unscaled). The strongest band of formamide and acetic acid is the C=O stretch near 1850–1870 cm^{-1}; phenol shows its O–H stretch near 3800 cm^{-1} and ring and C–O modes between 1150 and 1700 cm^{-1}. Experimental values are lower, which scaling corrects.

13. Step 6 — Single point and global reactivity (Part 9)

13.1 Accurate energies and frontier orbitals

What it does

On each optimised geometry the notebook runs one more calculation (a **single point**) with the larger basis set. It records the total energy, dipole moment and orbital energies, and, if Fukui analysis is on, also computes the molecule with one electron added (anion, N+1) and one removed (cation, N−1) at the same geometry.

Why it is needed

Orbital energies, dipoles and densities converge slowly with basis-set size, so they are computed with the larger basis even though the geometry came from the smaller one (Section 3.8).

The **frontier molecular orbitals** govern chemical reactivity. The **HOMO** (highest occupied molecular orbital) holds the most easily donated electrons; the **LUMO** (lowest unoccupied molecular orbital) is where incoming electrons go. Their energy difference, the **HOMO–LUMO gap**, is a classic index of kinetic stability: a small gap means a soft, polarisable, more reactive molecule.

What you get

- HOMO, LUMO and gap in eV; dipole moment; an orbital energy diagram and density of states (Figure 7).
- Conceptual-DFT descriptors, computed two ways (table below).

What to check

- Compare molecules within one study, at one level of theory.
- Prefer the ΔSCF values when they differ strongly from the orbital-based (Koopmans) values, especially for the electron affinity A.

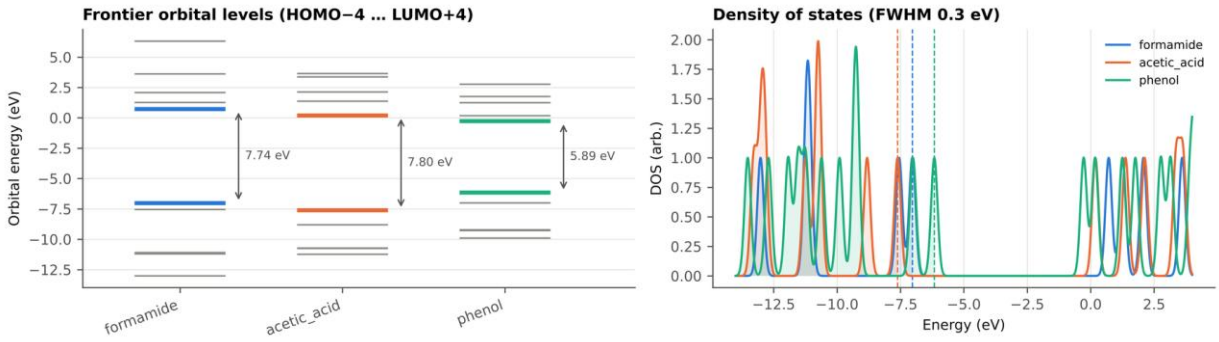


Figure 7. Frontier orbital energy levels with the HOMO–LUMO gap (left) and the density of states (right). Phenol's aromatic π system gives it the smallest gap of the three test molecules.

13.2 Conceptual-DFT reactivity descriptors

Conceptual DFT turns two numbers, the ionisation energy I (cost of removing an electron) and the electron affinity A (energy gained by adding one), into intuitive chemical indices. The notebook computes I and A in two ways: from orbital energies (Koopmans' approximation: I ≈ −E(HOMO), A ≈ −E(LUMO)), which most papers use, and from the actual energies of the N−1 and N+1 electron systems (ΔSCF), which is better founded.

Descriptor	Formula	Chemical meaning
Electronegativity χ	(I + A)/2	Tendency to attract electrons
Chemical potential μ	−χ	Direction of electron flow: electrons move from high μ to low μ
Chemical hardness η	(I − A)/2	Resistance to change of electron cloud; hard = stable, unreactive
Softness S	1/(2η)	Inverse of hardness; soft = polarisable, reactive
Electrophilicity ω	$\mu^2/(2\eta)$	Energy lowering on accepting electrons; high ω = strong electrophile
Max. electron uptake ΔNmax	−μ/η	How many electrons the molecule can accept
Electroaccepting power ω^+	$(I + 3A)^2/[16(I - A)]$	Ability to accept charge
Electrodonating power ω^-	$(3I + A)^2/[16(I - A)]$	Ability to donate charge; used to rank antioxidants

WATCH OUT

Some papers define hardness as η = I − A, without the factor ½. Both conventions are in use, so always state which one you used. DFT-Flow uses the Parr–Pearson definition, η = (I − A)/2.

14. Step 7 — Atomic charges and Fukui functions (Part 10)

14.1 Finding the reactive atoms

What it does

The notebook divides the electron density among the atoms to give **partial atomic charges**, using Hirshfeld partitioning (with Mulliken charges for comparison). It then computes **condensed Fukui functions**, which measure how much each atom's electron population changes when the molecule gains or loses an electron:

f^+ (from N → N+1): where an added electron goes, so where a **nucleophile** will attack. **f^-** (from N−1 → N): where an electron is removed from, so where an **electrophile** or oxidant will attack. **f^0**: their average, for radical attack. The **dual descriptor** $\Delta f = f^+ - f^-$ combines both: $\Delta f > 0$ marks an electrophilic site, $\Delta f < 0$ a nucleophilic site.

Why it is needed

Global descriptors describe the whole molecule; drug design usually needs to know **which atom** reacts. Fukui functions point to sites of metabolic oxidation, covalent reactivity (for example Michael acceptors), hydrogen-bond donors and acceptors, and positions for chemical modification.

Partial charges are not physically observable, and different schemes give different numbers. Hirshfeld charges depend very little on the basis set, which is why they are the standard choice for condensed Fukui functions; Mulliken charges are basis-set sensitive but appear in older literature.

What you get

- Per-atom tables of charges and Fukui indices for each molecule.
- A 2D map coloured by the dual descriptor and a bar chart of f^+ and f^- per atom (Figure 8).
- A summary table of the top sites for each kind of attack.

What to check

- Each Fukui function sums to 1.00 over all atoms (printed by the notebook).
- Results agree with chemical intuition, for example carbonyl carbons as electrophilic sites.

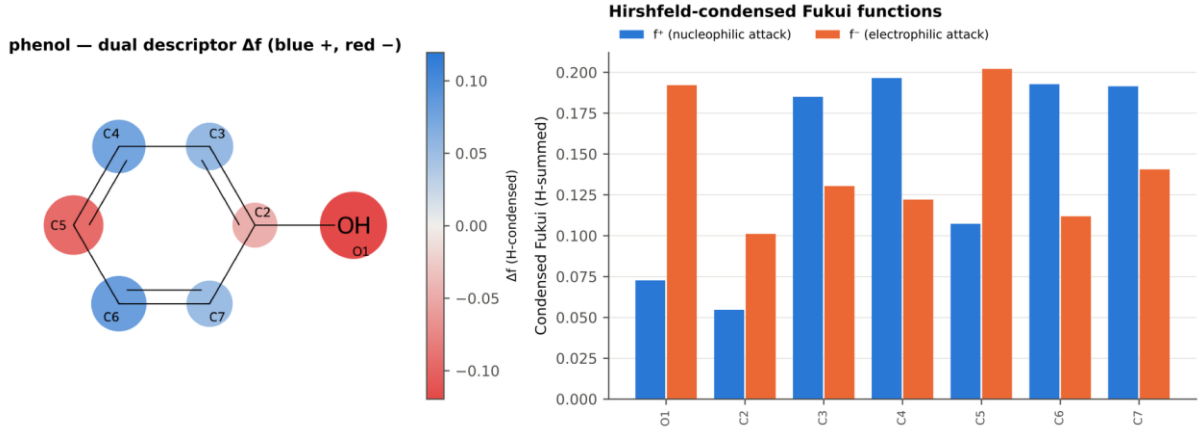


Figure 8. Fukui analysis of phenol. Left: dual descriptor (red = nucleophilic, attacked by electrophiles; blue = electrophilic). Right: f^+ and f^- per heavy atom. The largest f^- is on the para carbon (C5) and the oxygen, reproducing the textbook ortho/para-directing effect of the OH group in electrophilic substitution.

15. Step 8 — Molecular electrostatic potential and orbital surfaces (Part 11)

15.1 Visualising charge and orbitals in 3D

What it does

The **molecular electrostatic potential (MEP)** is the electrostatic energy a positive test charge would feel at each point around the molecule, from the nuclei and the electron cloud together. The notebook evaluates it exactly on the molecule's surface, defined as the contour where the electron density equals 0.001 atomic units (approximately the van der Waals surface), and colours it: **red** for negative (electron-rich) regions and **blue** for positive (electron-poor) regions. It also draws the HOMO and LUMO as 3D lobes, with the two colours showing the two phases (signs) of the orbital.

Why it is needed

Molecular recognition is largely electrostatic. Red regions such as carbonyl oxygens and nitrogen lone pairs are hydrogen-bond acceptors and attract cations or positively charged residues; blue regions such as O–H and N–H hydrogens are hydrogen-bond donors. Comparing the MEP of a ligand with its docking pose shows whether its interactions make electrostatic sense.

The orbital pictures show **where** the frontier electrons are located. They should agree with the f^- (HOMO) and f^+ (LUMO) maps from the previous step.

What you get

- Publication renders of the MEP, HOMO and LUMO (Figure 9) and an interactive, rotatable 3D viewer.
- **Murray–Politzer surface descriptors**: the most negative and most positive surface values ($V_{S,min}$ and $V_{S,max}$) with the nearest atom, average positive and negative potentials, local polarity Π, variances σ^2, balance ν, surface area and molecular volume. These are used as descriptors in QSAR and solubility models.
- Cube files that open in VMD, Avogadro, Chimera, GaussView or Multiwfn for custom images.

What to check

- $V_{S,min}$ sits on the atom you expect (e.g. a carbonyl or hydroxyl oxygen).
- Colour scales are symmetric about zero; compare molecules using the numbers in the descriptor table rather than colours alone, because each figure is scaled to its own range.

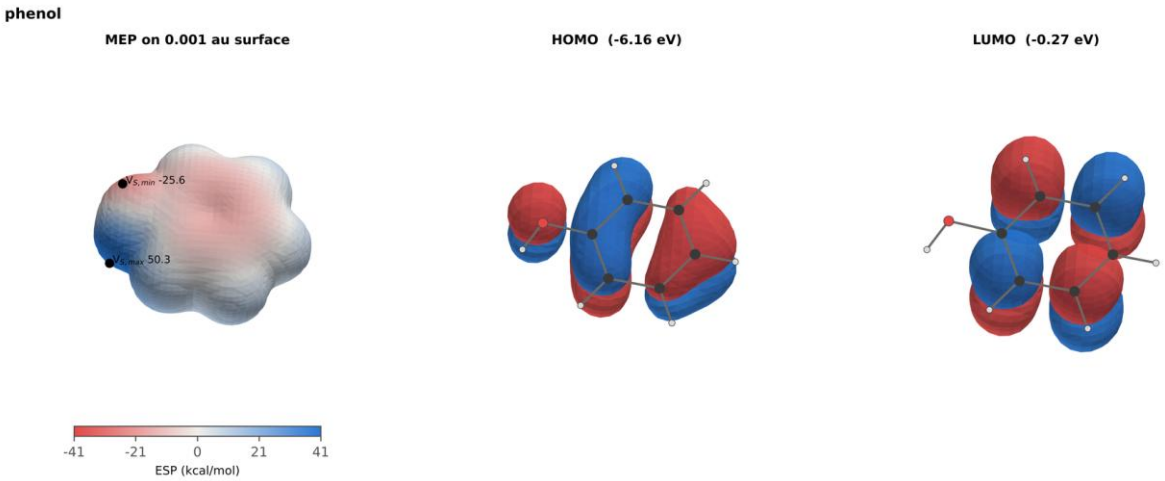


Figure 9. Phenol: MEP on the 0.001 au surface (left), with the most negative region at the oxygen lone pairs and the most positive at the hydroxyl hydrogen; HOMO (middle) and LUMO (right), both π orbitals of the ring, shown at ±0.03 au.

16. Step 9 — Excited states and UV-Vis spectra (Part 12)

16.1 Time-dependent DFT

What it does

Ordinary DFT describes the ground state. **Time-dependent DFT (TD-DFT)** computes how the molecule responds to light, giving the energies of the lowest excited states (**vertical excitation energies**, at the ground-state geometry) and their **oscillator strengths** f, which measure how intense each absorption is. The notebook converts these into a simulated UV-Vis spectrum by giving each transition a Gaussian band shape, and lists which orbital promotions (for example HOMO → LUMO) make up each state.

Why it is needed

UV-Vis spectra are cheap to measure, and TD-DFT explains them: which band corresponds to which electronic transition, and how substituents shift the absorption. This matters for dyes, photosensitisers, natural pigments, and for confirming structures of isolated compounds.

The functional matters here more than anywhere else. Range-separated functionals such as CAM-B3LYP (the notebook's default) describe charge-transfer transitions correctly, whereas B3LYP can place them far too low in energy.

What you get

- Simulated spectrum with λmax labelled, a stick spectrum of oscillator strengths (Figure 10), and a table of every state with its energy, wavelength, f and orbital character.

What to check

- Typical accuracy is about 0.2–0.4 eV. Gas-phase calculations with small basis sets usually predict absorption at shorter wavelengths than measured in solution; use the matching solvent and the Standard preset before comparing with experiment.
- Enough states: if the notebook reports that the computed states do not reach the plot window, increase the number of states.
- States with f ≈ 0 are “dark” and do not appear in the spectrum, but they still exist.

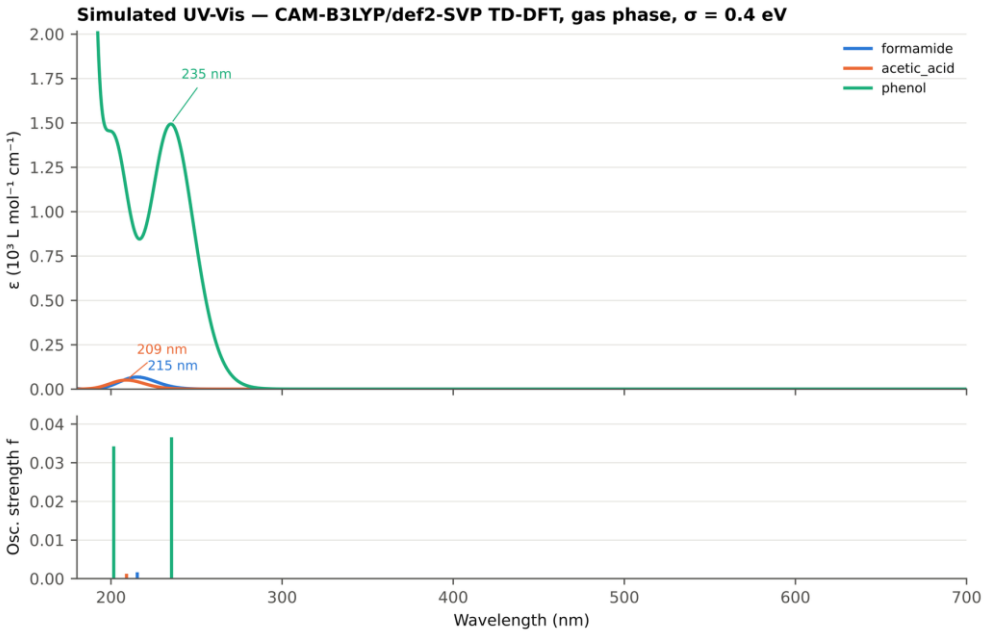


Figure 10. Simulated UV-Vis spectra from the test run (CAM-B3LYP/def2-SVP, gas phase). Phenol's first strong band is computed at 235 nm; experimentally it lies near 270 nm in solution. This blue shift is typical of a small-basis gas-phase calculation and illustrates why the protocol should be validated before comparing with experiment.

17. Step 10 — Comparison, export and reporting (Parts 13–14)

17.1 Bringing the results together

What it does

The notebook merges every result into one master table, highlights the extreme molecule for each key property (smallest gap, strongest electrophile, most polar and so on), and draws comparison plots (Figure 11). It then exports structures and data, writes a methods paragraph with numbered references matched to exactly what was run, builds a PDF report, and packages everything into a ZIP file.

Why it is needed

A study usually compares a series: analogues of a lead, phytochemicals from one plant, designed versus reference compounds. Standardised comparison, and a methods section that exactly matches the calculation, make the results usable in a manuscript and reproducible by others.

What you get

- Master summary table (CSV and Excel), comparison figures, PDF report, methods text with references, ZIP archive.
- An SDF file of DFT-optimised structures with properties attached, ready for docking, pharmacophore or QSAR work in other BinFoLab notebooks.
- Molden files for viewing orbitals in Avogadro, Jmol or Chemcraft.

What to check

- Read the auto-generated methods text once against your journal's style.
- Treat the automatic highlights as observations to compare with your biological data, not as conclusions.

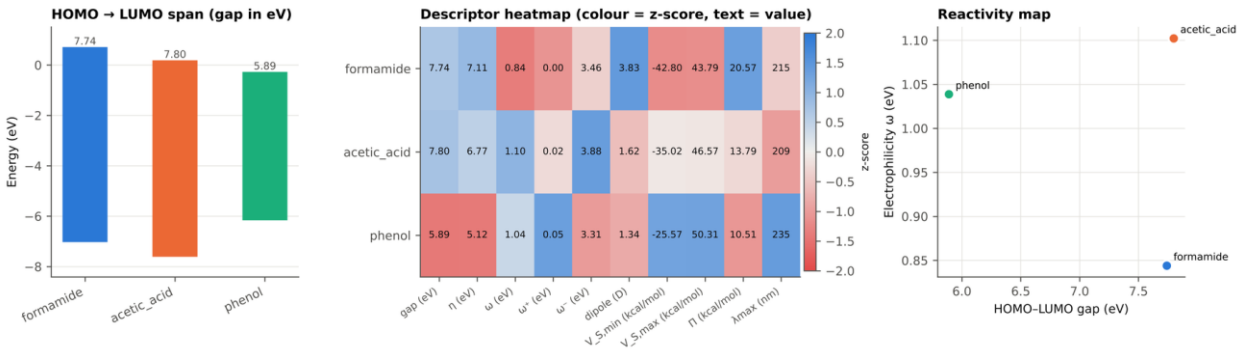


Figure 11. Cross-molecule comparison: HOMO→LUMO span for each molecule (left), a heatmap of standardised descriptors with raw values printed (middle), and a reactivity map of gap against electrophilicity (right).

17.2 What each output file is for

File or folder	Contents	Use it for
tables/10_master_summary.csv	One row per molecule, all key results	The main results table of your paper
tables/DFTFlow_results.xlsx	Every table as a separate sheet	Supporting information

File or folder	Contents	Use it for
structures/DFT_optimised_structures.sdf	Optimised 3D structures with properties	Docking, pharmacophore, QSAR
structures/*_dft_opt.xyz	Optimised coordinates	Supporting information (journals often require them)
figures/	All figures at 600 DPI (PNG) and as vector SVG	Manuscript figures
cubes/, molden/	Density, orbital and ESP grids; orbital files	Custom images in VMD, Avogadro, Chimera
report/methods.md, report/DFTFlow_report.pdf	Methods paragraph, references and complete report	Methods section; record keeping

18. Interpreting results: a worked example

The test molecule phenol shows how the separate outputs fit into one consistent picture. A good DFT discussion connects them in the same way.

1. **Global reactivity.** Phenol has the smallest HOMO–LUMO gap of the three test molecules (5.9 eV versus about 7.8 eV for formamide and acetic acid), so it is the softest and most polarisable. Its computed ionisation energy of 8.38 eV is close to the experimental 8.49 eV, which gives confidence in the method.
2. **Where electrons are donated from.** The HOMO is a π orbital spread over the ring and the oxygen (Figure 9), and the largest f^- values sit on the para carbon and the oxygen (Figure 8).
3. **Chemical consequence.** Together these predict that electrophiles attack the ring at the para (and ortho) positions, exactly the ortho/para-directing behaviour of the OH group known from organic chemistry, and that the O–H group is a reactive site for oxidation, consistent with phenols acting as antioxidants.
4. **Interactions.** The MEP shows a negative region at the oxygen (a hydrogen-bond acceptor) and the most positive region at the hydroxyl hydrogen (a hydrogen-bond donor), so phenol can both donate and accept hydrogen bonds in a binding pocket.

> **BEGINNER TIP**
> A strong DFT discussion does not list numbers. It asks whether the global descriptors, the local reactive sites, the MEP and the spectra tell the **same story**, and whether that story agrees with experiment or biological activity.

18.1 Quick interpretation reference

Result	Higher / larger value means…
HOMO energy (less negative)	Better electron donor; easier to oxidise
LUMO energy (more negative)	Better electron acceptor; easier to reduce
HOMO–LUMO gap	More kinetically stable, harder, less reactive
Electrophilicity ω	Stronger electrophile (empirical scale: > 1.5 eV strong, 0.8–1.5 moderate, < 0.8 marginal)
Dipole moment	More polar; generally more water-soluble
Atom with high f^-	Likely site of electrophilic attack or oxidation
Atom with high f^+	Likely site of nucleophilic attack
Very negative $V_{S,min}$	Strong hydrogen-bond acceptor site
Very positive $V_{S,max}$	Strong hydrogen-bond donor site
Longer λmax	Absorbs at lower energy (more extended conjugation)

19. Reporting DFT in a manuscript

Reviewers need to be able to reproduce your calculations. A complete computational section states:

- Software and version (e.g. PySCF 2.14 with GPU4PySCF).
- How starting structures and conformers were obtained.
- Functional, dispersion correction and basis set for **both** optimisation and single point, using the notation of Section 3.8.
- Solvent model and solvent, or that calculations were in the gas phase.
- That frequency calculations confirmed every structure is a minimum, the temperature for thermochemistry, and any frequency scale factor.
- Definitions of every descriptor (especially the hardness convention) and the charge scheme used for Fukui functions.
- For TD-DFT: functional, number of states, and the broadening used for spectra.
- Optimised Cartesian coordinates in the supporting information.
- References to the original method papers. DFT-Flow generates this paragraph and reference list automatically (Part 14.2).

20. Common beginner mistakes

Mistake	Consequence	How to avoid it
Wrong protonation state or charge	A different molecule is calculated; all results are wrong	Supply the intended microspecies (e.g. at pH 7.4); check the input table
Not checking for imaginary frequencies	A saddle point is reported as a stable structure	Always run frequencies; DFT-Flow checks and repairs automatically
Comparing numbers from different functionals or basis sets	Differences reflect the method, not the molecules	Use one level of theory for the whole study
Treating the HOMO–LUMO gap as the optical gap	Orbital gaps from hybrid DFT do not equal absorption energies	Use TD-DFT for UV-Vis comparisons
Using only one conformer of a flexible molecule without checking	Properties may not represent the real molecule	Inspect conformer populations (Section 10)
Mulliken charges with large or diffuse basis sets	Unphysical charges that change with the basis	Use Hirshfeld (default in DFT-Flow) or ESP-derived charges
Over-interpreting small differences	Claims not supported by method accuracy	Treat differences of a few hundredths of an eV as insignificant
Claiming DFT proves biological activity	Unsupported conclusions in review	Present DFT as explanatory support alongside docking, MD and experiment

21. Glossary, units and further reading

21.1 Glossary

Term	Meaning
Basis set	The set of atom-centred functions used to build molecular orbitals
Conformer	One of the 3D shapes a molecule can adopt by rotation about single bonds
Converged	An iterative calculation (SCF or optimisation) whose results no longer change within set thresholds
Density fitting (RI)	A numerical approximation that speeds up the calculation of electron repulsion with negligible loss of accuracy
Dispersion correction	An added energy term (D3, D4) for van der Waals attraction, which plain DFT misses
Electron density ρ(r)	Probability of finding an electron at point r; the central quantity of DFT
Functional	A rule that maps the density to an energy; in practice, the exchange–correlation approximation
Hessian	Matrix of second derivatives of energy with respect to atomic positions; gives vibrational frequencies
HOMO / LUMO	Highest occupied / lowest unoccupied molecular orbital
Imaginary frequency	A vibration with negative curvature; indicates a saddle point, not a minimum
Isosurface	A 3D surface on which a quantity (density, orbital) has a constant value
Level of theory	The combination of method, functional, basis set, dispersion and solvent
Multiplicity	2S+1, where S is total spin: 1 = singlet (paired), 2 = doublet (one unpaired electron)
SCF	Self-consistent field: the iterative procedure that finds the converged orbitals and density
Single point	A calculation at a fixed geometry, without optimisation
Vertical excitation	Electronic excitation at the ground-state geometry, as in light absorption

21.2 Unit conversions

Quantity	Conversion
Energy	1 hartree (Eh) = 27.2114 eV = 627.51 kcal/mol = 2625.5 kJ/mol
Energy	1 eV = 23.06 kcal/mol = 96.49 kJ/mol
Wavelength from energy	λ (nm) = 1239.84 / E (eV)
Length	1 bohr = 0.52918 Å
Dipole moment	1 atomic unit = 2.5417 debye

21.3 Further reading

- Koch, W.; Holthausen, M. C. *A Chemist's Guide to Density Functional Theory*, 2nd ed.; Wiley-VCH, 2001. The most accessible book-length introduction for chemists.
- Cramer, C. J. *Essentials of Computational Chemistry: Theories and Models*, 2nd ed.; Wiley, 2004.
- Jensen, F. *Introduction to Computational Chemistry*, 3rd ed.; Wiley, 2017.
- Bursch, M.; Mewes, J.-M.; Hansen, A.; Grimme, S. Best-practice DFT protocols for basic molecular computational chemistry. *Angew. Chem. Int. Ed.* 2022, 61, e202205735. Practical recommendations on functionals, basis sets and workflows.
- Geerlings, P.; De Proft, F.; Langenaeker, W. Conceptual density functional theory. *Chem. Rev.* 2003, 103, 1793–1874. The reference review for reactivity descriptors and Fukui functions.
- Hohenberg, P.; Kohn, W. Inhomogeneous electron gas. *Phys. Rev.* 1964, 136, B864–B871.
- Kohn, W.; Sham, L. J. Self-consistent equations including exchange and correlation effects. *Phys. Rev.* 1965, 140, A1133–A1138.
- Perdew, J. P.; Schmidt, K. Jacob's ladder of density functional approximations for the exchange-correlation energy. *AIP Conf. Proc.* 2001, 577, 1–20.
- Sun, Q. et al. Recent developments in the PySCF program package. *J. Chem. Phys.* 2020, 153, 024109.

The complete list of method references used by the pipeline is generated by DFT-Flow in report/methods.md.